Lesson notes · for you

# Fingerprints of data: hashing

Cyber security · Level 4 · Lesson 1

**Time**
17 min

**Materials**
A phone or computer for the lab (one per pair is enough); Paper and coloured pens

## Learning goals

- Describe the three rules of a good hash.
- Explain how websites store passwords as fingerprints.
- Use fingerprints to check that a file wasn't changed.

## Activity: Fingerprint detective

1. Warm-up (3 min): Answer the puzzle in your own words: how can a website check your password without keeping it?
2. Lab (6 min): In the "Fingerprints of data" lab, make the fingerprint of a sentence. Change one letter, then note how many of the 256 bits change and how many characters look different.
3. Detect (5 min): Your partner hashes two short sentences and writes down the first eight characters of each fingerprint, then secretly changes one sentence by a single letter. Hash both again and find which one changed.
4. Explain (3 min): Tell your partner why a site that can email you your password is a warning sign.

**Work in pairs with one phone or computer. Before you start, each of you writes two short sentences on paper.**

## Discussion

- Why does a good website never need to know your password?
- What would go wrong if a hash could be run backwards?
- Why does a long passphrase still matter if the site uses hashes?

## Check yourself

- One use of a hash, and why it works.

**Try it this week**
Hash your name in the lab, change one letter and compare the two fingerprints. Then find the fingerprint (often labelled SHA-256) next to a download on an official website, and explain to a friend or someone at home what it is for.

Worksheet

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Name _____ Class _____ Date _____

**1** **A website emails you your old password "in case you forgot it". What does that tell you?**

- ◯ It stores your password itself, not a fingerprint
- ◯ It's taking good care of your password
- ◯ It uses an extra-strong hash

**2** **If you know a SHA-256 fingerprint, you can run the hash backwards to find the password it came from.**

- ◯ True
- ◯ False

**3** **Which of these are good jobs for a hash?**

- ☐ Checking a downloaded file wasn't changed
- ☐ Storing passwords so the site never keeps them
- ☐ Shrinking a photo to save space on your phone
- ☐ Checking a backup copy is exact
- ☐ Hiding a message so a friend can read it later

**4** **Complete the three rules of a good hash.**

The same input always gives _____ ; change one letter and about _____ of its bits change; and it only works _____ .

**Word bank:** half · both ways · a quarter · the same fingerprint · one way · a new fingerprint

✎ **Which of your accounts would hurt most if that site's password fingerprints leaked? What makes your password there hard to guess, even for someone with its fingerprint?**

Check your answers

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1 **It stores your password itself, not a fingerprint**

A hash can't be run backwards, so a site that kept only the fingerprint could never send your password back.

2 **False**

A hash is one-way: there is no way back from the fingerprint.

3 **Checking a downloaded file wasn't changed; Storing passwords so the site never keeps them; Checking a backup copy is exact**

Hashes are for checking: matching fingerprints prove a download, a password or a backup is exactly right.

4 **The same input always gives the same fingerprint; change one letter and about half of its bits change; and it only works one way.**

Same input, same fingerprint, so sites can compare.